Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3471-WE01

Title: **MEng/BEng:** Materials 3		
Time Allowed:	2 hours	
Additional Material provided:	Yes	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Section A
Question 1

a) A 1m long steel rod (2 cm in diameter) is coated with a 0.5-cm-thick layer of zinc. A current density of 0.0002 A/cm^2 is applied to the rod in an aqueous solution. Assuming that the zinc corrodes uniformly, determine the length of time required before the steel is exposed. The molecular weight of Fe, C and Zn are 55.84, 12 and 65.38 g/mol, respectively. Densities of Fe, C and Zn are 7.87, 2.26 and 7.13 g/cm^3, respectively. Faraday's constant is 96500 C, and the "n" of zinc and iron ions are 2.

[20%]

b) In view of the phase diagram shown below (cf. Figure Q1), we say that copper can contain up to 40% Zn and still be single phase. How do we explain this statement? In general, under which conditions can two elements show unlimited solid solubility?

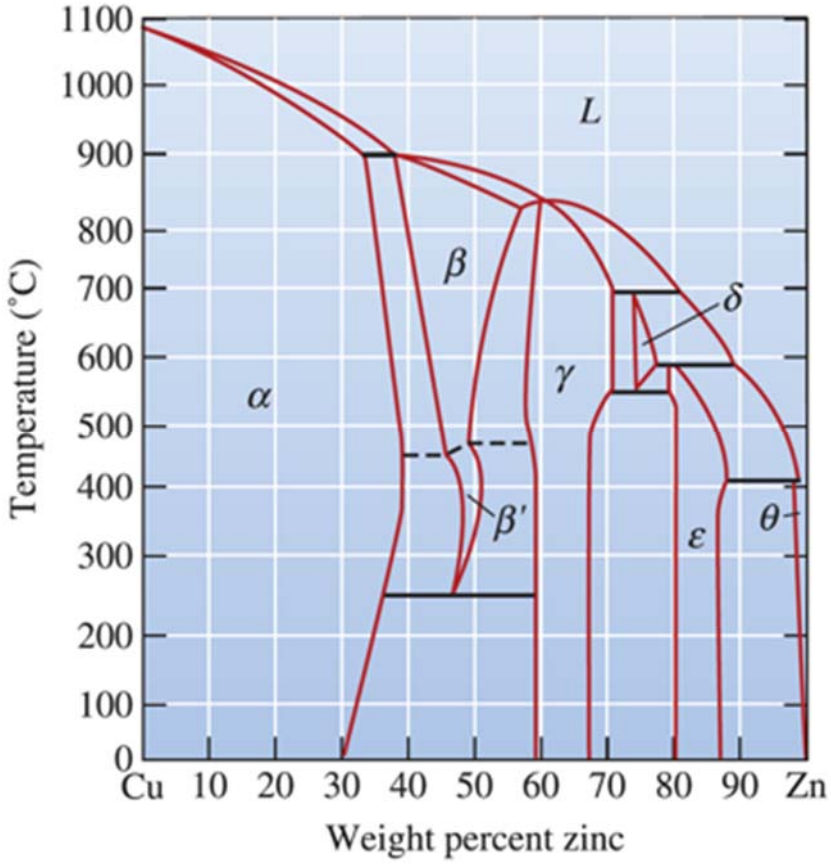


Figure Q1: Phase digram of Cu-Zn.

[20%]

c) Determine the interplanar spacing and the length of the Burgers vector for slip on the expected slip systems in BCC tantalum. Lattice parameter of tantalum is 0.3305 nm.

[30%]

d) For an ASTM grain size number of 9, calculate the number of grains per square inch with no magnification.

[15%]

e) The leaf springs for an automobile are formed from a high-carbon steel. For best corrosion resistance, should the springs be formed by hot working or cold working? Explain.

[15%]

continued

Section A
Question 2

a) Silicon carbide (SiC) has a specific gravity (true density) of 3.1 g/cm^3. A sintered SiC part is produced, occupying a volume of 500 cm^3 and weighing 1200 g. After soaking in water, the part weighs 1250 g. Calculate the bulk density, the true porosity, and the volume fraction of the total porosity that consists of closed pores.

[25%]

b) How many grams of BaO can be added to 1 kg of SiO_2 before the O:Si ratio exceeds 2.5 and glass-forming tendencies are poor? Molecular weights of Ba, Si and O are 137.3, 28.1 and 16 g/mol, respectively.

[25%]

c) A polymer component that needs to maintain a stress level above 10 MPa for proper functioning of an assembly is scheduled to be replaced every two years as preventative maintenance. If the initial stress was 18 MPa and it dropped to 15 MPa after one year of operation (assuming constant strain), will the part replacement under the specified preventative schedule successfully prevent a failure?

[20%]

d) Analysis of a sample of polyacrylonitrile ($(C_3H_3N)_n$) shows that there are six lengths of chains.The number and mean molecular weight of chains are given in the Table Q2. Molecular weights of C, H and N are 12, 1 and 14, respectively.

Determine
I. the weight average molecular weight and degree of polymerization, and
II. the number average molecular weight and degree of polymerization.

[30%]

Table Q2: Number of Chains and Mean Molecular Weight of chains of polyacrylonitrile.

Number of Chains	Mean Molecular Weight of Chains (g/mol)
10,000	3,000
18,000	6,000
17,000	9,000
15,000	12,000
9,000	15,000
4,000	18,000

continued

Section B
Question 3

a) Give a definition of sustainable development including the words economy, society and environment.

[10%]

b) Use one example to explain what an externality (to our industrial economy) is.

[20%]

c) Explain why externalities are intensified by climate change using social, economic and environmental examples.

[30%]

d) Give one example of international guidance and one example of UK legislation which are helping us to internalise the externalities into our industrial economy.

[40%]

Section B
Question 4

The floor of a hut for an arctic survey team sits on a set of short posts which in turn sit on "foundation boards" on the snow, as illustrated in Figure Q4. The posts need to support the weight of the floor (60 kN per post including factors of safety) while transferring as little heat as possible from the floor of the hut to the foundation boards.

Table QT4 provides material properties and selected material indices. Please answer this question based on properties in this table.

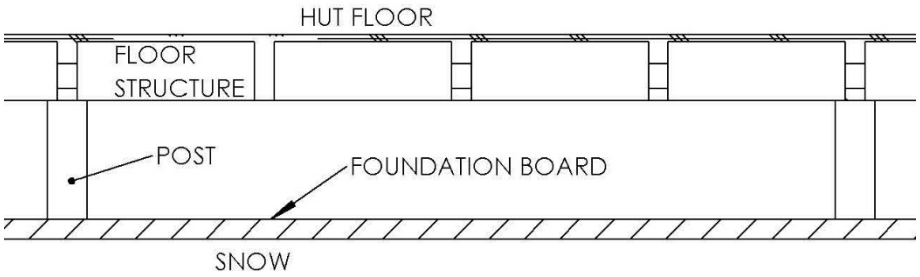


Figure Q4 – Section through floor assembly – Not to Scale

a) Derive an appropriate material index for the posts and justify your assumed failure mechanism.

[50%]

b) Identify 6 materials which are attractive in terms of the material index.

Recommend a material, justifying your selection.

Sketch a suitable cross section for the material chosen explaining your rationale for the geometry chosen.

[50%]

Table QT4 – Material Properties and Indices

Material	E Young's Modulus (GPa)	σ_f Failure Stress (MPa)	k Thermal Conductivity (W/mK)	$\frac{E}{k} \times 10^{-9}$ $\left(\frac{K\,s}{m^3}\right)$	$\frac{E^{\frac{1}{2}}}{k} \times 10^{-3}$ $\left(\frac{Ks^2}{kg^{\frac{1}{2}}m^{\frac{3}{2}}}\right)$	$\frac{\sigma_f}{k} \times 10^{-6}$ $\left(\frac{K\,s}{m^3}\right)$	$\frac{\sigma_f^{2/3}}{k} \times 10^{-3}$ $\left(\frac{Ks^{\frac{5}{3}}}{kg^{\frac{1}{3}}m^{\frac{5}{3}}}\right)$
ABS	2.00	35.0	0.265	14.6	86.9	132.1	403.8
Aluminium Alloy 6083T6	70.0	240	158	27.4	21.0	1.5	2.4
CFRP	55.1	483	1.95	26.8	168.1	247.7	315.7
Concrete	20.0	23.0	1.60	5.9	111.8	14.4	50.5
Copper Alloy	130	265	275	15.3	21.7	1.0	1.5
GFRP	21.5	150	0.475	25.6	212.8	315.8	594.3
Grey Cast Iron	110	280	56.0	25.9	44.3	5.0	7.6
Low Alloy Steel	215	950	44.5	8.7	69.5	21.3	21.7
Low Carbon Steel	210	325	51.5	11.3	63.9	6.3	9.2
Magnesium Alloy	44.5	220	105	24.1	20.6	2.1	3.5
Natural Rubber	0.00200	25.0	0.120	35.5	4.1	208.3	712.5
Nylon	2.90	73.0	0.240	2.6	109.9	304.2	727.8
Polypropelene	1.25	29.0	0.140	1.8	94.5	207.1	674.2
Rigid Polymer Foam (HD)	0.340	7.00	0.0485	0.002	83.7	144.3	754.5
Tin	43.0	11.0	61.0	1.4	26.6	0.2	0.8
Titanium Alloy 6AL4v	114	880	10.5	18.6	104.2	83.8	87.5
Wood (along grain)	13.0	50.0	0.345	0.5	194.1	144.9	393.4

continued

Section B
Question 5

a) Metal powder is a finely divided particulate solid. Describe with the aid of diagrams the different particle shapes of metal powder.

[30%]

b) Welding is a fabrication process whereby two or more parts are fused together by means of heat, pressure, or both forming a join as the parts cool. Fusion welding is a welding process that uses heat to melt the base metals. Use diagrams to explain the three different types of fusion welding.

[50%]

c) The welding power generated in a particular arc-welding operation is 4000W. This is transferred to the work surface with a heat transfer factor of 0.70. The metal to be welded is Magnesium, whose melting point is 940 K. Assume that the melting factor = 0.2. A continuous fillet weld is to be made with a cross-sectional area = 25.00 mm^2. Determine the travel speed at which the welding operation can be accomplished.

[20%]

END